SAP Analytics Cloud - Modeler

This case study explains the creation process of an analytical model in SAP Analytics Cloud (SAC): from the initial creation and maintenance of properties, to the maintenance of dimensions and measures, to the loading and testing of data.

Product

SAP Analytics Cloud
Global Bike

Level

Bachelor's Degree
Master
Beginner

Focus

Business Intelligence
Data Modeling

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Version

3.0

Last Change

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MOTIVATION

An example sales dashboard was demonstrated in the exercises for SAP Analytics Cloud.

In the first case study, you developed a sales dashboard in the SAC Story Builder, but based on an existing model filled with data. In this second case study, you create your own model, load it with data, and evaluate it for testing with a simple story.

PREREQUISITES

Before you edit the case study, you should have familiarized yourself with the navigation in SAP Analytics Cloud. In addition, you should have successfully processed the SAC exercises and the first SAC case study on the Story Builder.

REMARK

This case study uses a model with historical sales data from the fictitious company Global Bike.

THANK YOU

This case study was developed with the help of students and employees of the Pforzheim University of Applied Sciences as well as scientific and student assistants of the SAP UCC Magdeburg.

SAP University Alliances

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Process Overview

Creating an Analytic Model in SAC **learning objective**.

Time 130 min.

Scenario Alex Kuhn (Vice President Marketing and Sales) wants to store a data record directly in the CLOUD solution *SAP Analytics Cloud (SAC)* to evaluate and research past data. Consistent figures from sales orders are defined in the ERP system for the period 1. January 2007 to 31 December 2018 and have already been extracted. The data record 2019 is currently being created and is to be reloaded. The task for Andrea Marino (Sales Manager USA) and Hanna Ortmann (Head of Sales Germany) is now to design the data model with her project team, load the data mentioned, and prepare it for subsequent analysis and visualization.

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Step 1: Log on to SAC

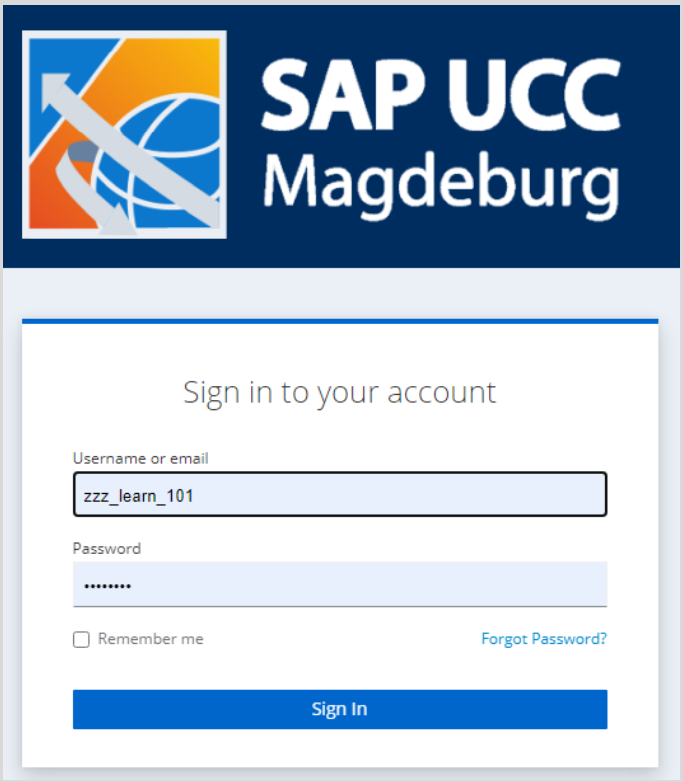
Task Log on to SAP Analytics Cloud. **Time** 5 min.

Description Log on to the SAC tenant in a web browser.

Initial Screen

Clicking on the following link takes you to the Identity Management System of SAP UCC Magdeburg. This provides anonymous identities for logging on to SAP cloud solutions according to valid data protection rights.

https://ucc-pilot-sac.eu10.hcs.cloud.sap/



User
Password

On the login screen, enter the **user** named by your instructor and your personal **password**, which you should have already defined.

Sign In

Then click **Sign In**.

Navigation Course

Note If you encounter problems when registering, please check the navigation course or consult your instructor.

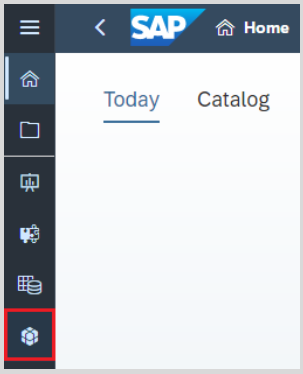
Step 2: Create Model

Task Create a new model. **Time** 15 min.

Description Use the modeler to start building your model.

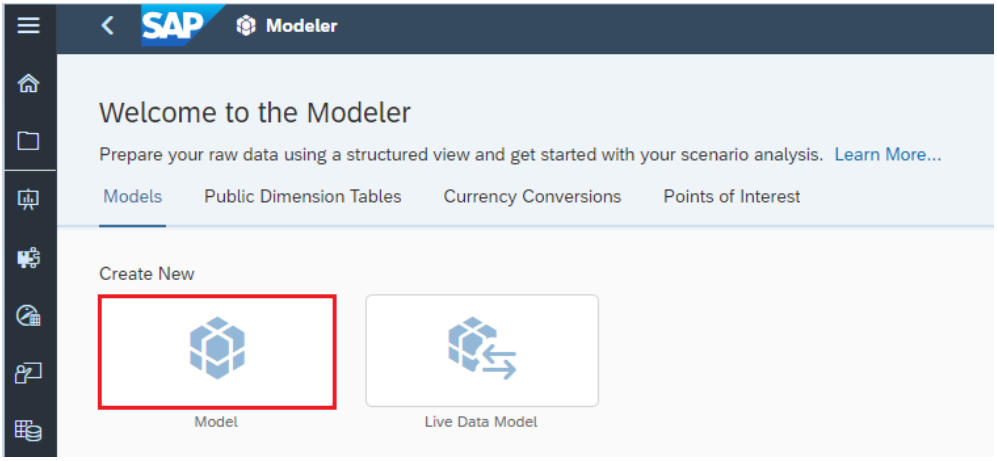
On the home screen, click Modeler in the upper left corner.

Initial Screen



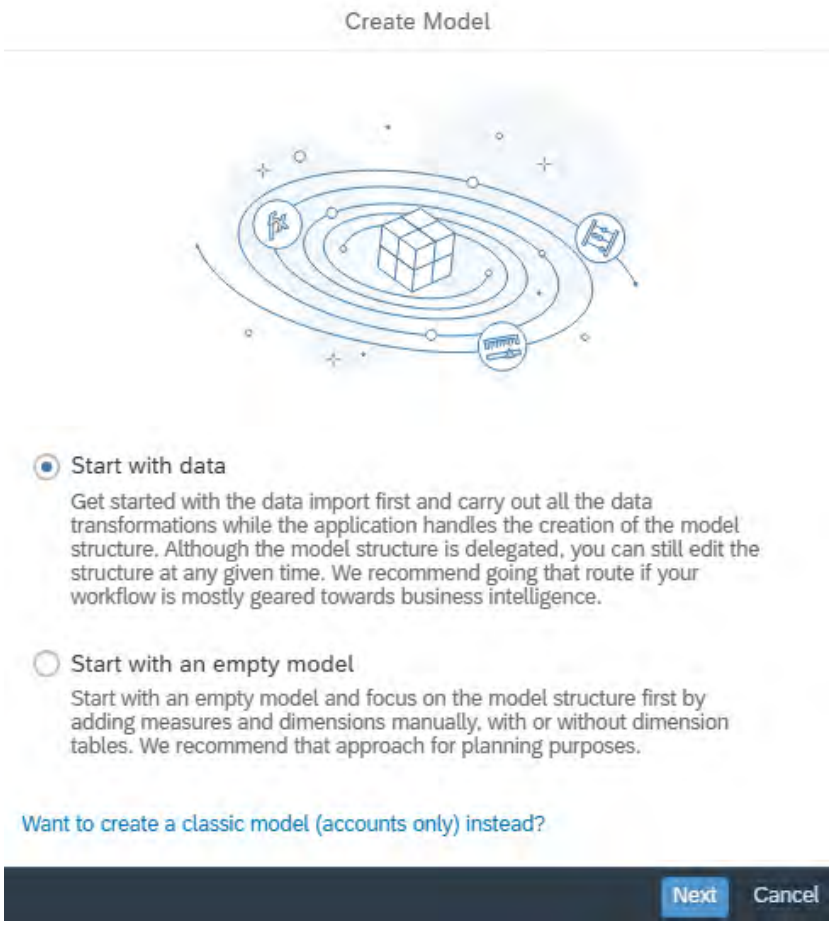
Click the **Model** (*Create New* Model) button in the upper left under *Create New*.

Model



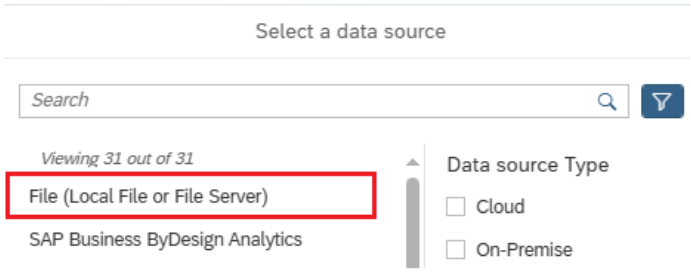
In the window that appears, choose **Start with data** and confirm your selection with **Next**.

Start with data
Next



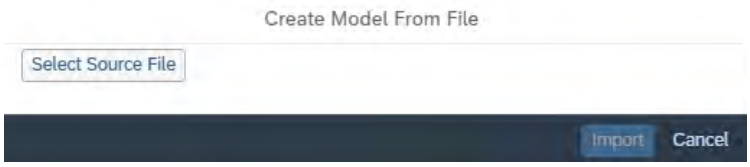
Under *Select a data source*, choose **File (Local File or File Server).**

File

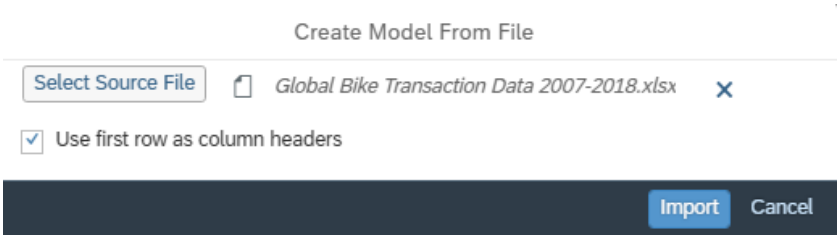


Click the **Select Source File** button to select your data.

Select Source File

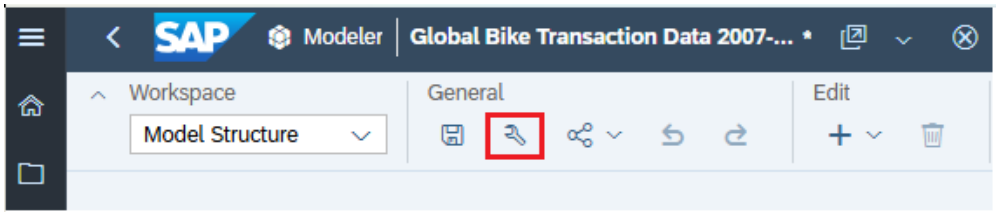


In the window that opens, search for the Excel file *Global Bike Transactional Data 2007-2018.xlsx* and choose **Import**.



The data is imported. You then make model preferences.

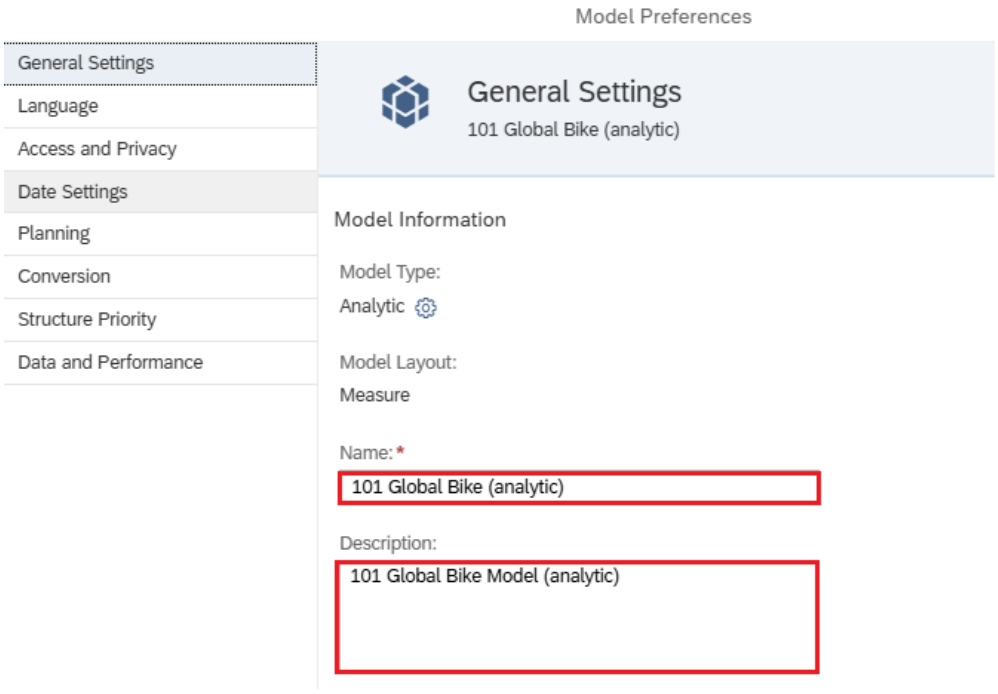
In the menu bar on the upper left, click  (*Model Preferences*).



Stay in the window that opens on the current tab (*General Settings*).

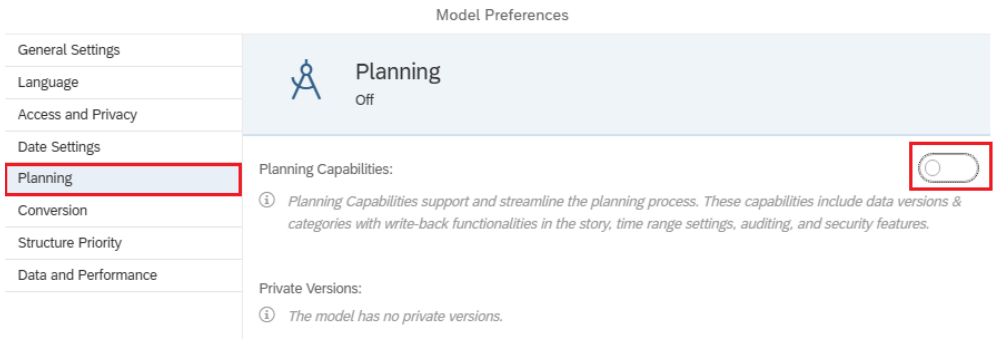
Then enter **### Global Bike (analytic)** in the *Name* field and **### Global Bike Model (analytic)** in the *Description* field. Note that you replace ### with your three-digit number provided by your instructor.

Global Bike (analytic)
Global Bike Model (analytic)



Change the type of your model to *Analytic*. To do this, go to the *Planning* tab and deactivate the switch for **Planning Capabilities**.

~~Planning Capabilities~~



Note After you have saved your model for the first time later, it is no longer possible to change the model type. Therefore, please make sure again that the switch is actually deactivated (colorless).

Confirm your changes with **OK**.

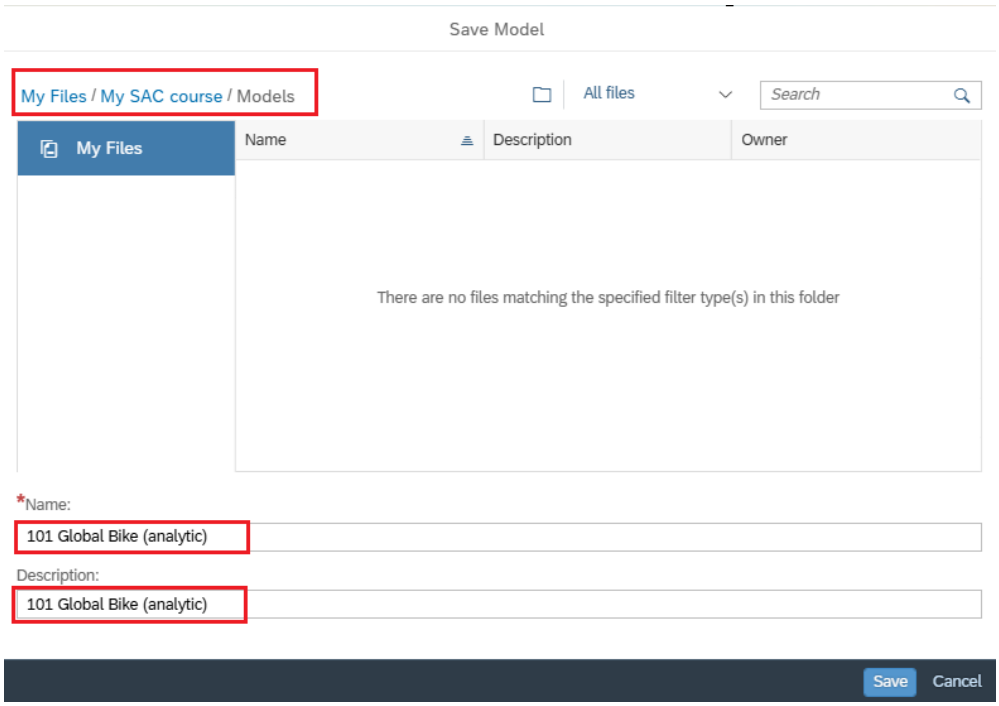
Ok then

Save your model. To do this, click **Save** icon in the *General* area of the ribbon.

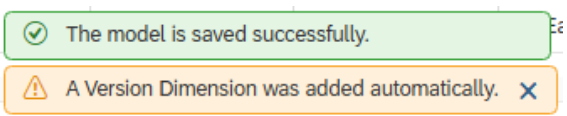
Save

In the window that opens, navigate to your My *SAC course* folder in the *My Files* folder and then to the *Models* subfolder. Check the name and description of your model again, especially if you have replaced ### with your three-digit number. Now click **Save** to save your model.

Save



The SAC system confirms the successful backup of your model with a success message.



Note Since planning is also possible in SAC, all models are automatically assigned the dimension *Version*.

Remain in this window for the next step.

☐

Step 3: Change Dimension Date

Task Change the *Calendar_Day* dimension. **Time** 5 min.

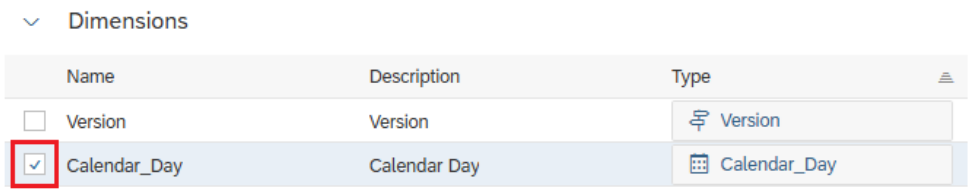
Description First, change the name and hierarchy of your time dimension to *Year*, *Month*, and *Day*.

Your *### Global Bike (analytic)* model should still be open. If not, restart the *Modeler* and open your model.

Initial Screen

Under *Dimensions*, select the **Calendar_Day dimension**.

Calendar_Day

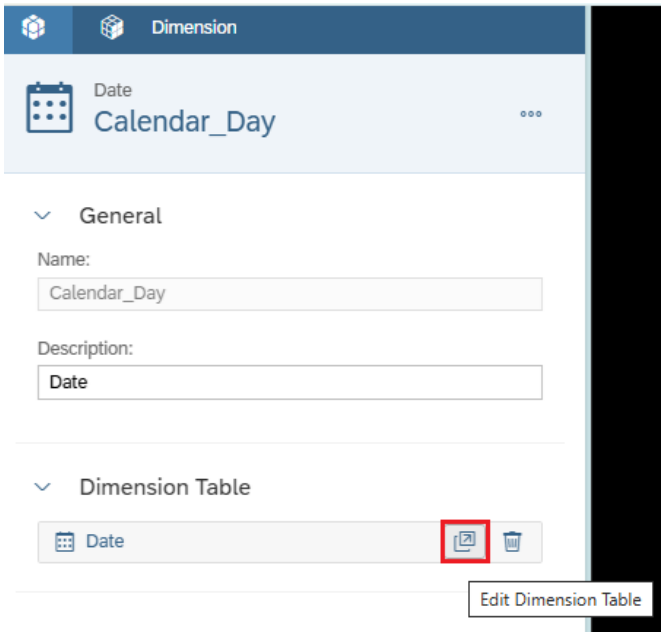


In the right pane, the *dimension* opens. Change the Description to **Date**.

Date

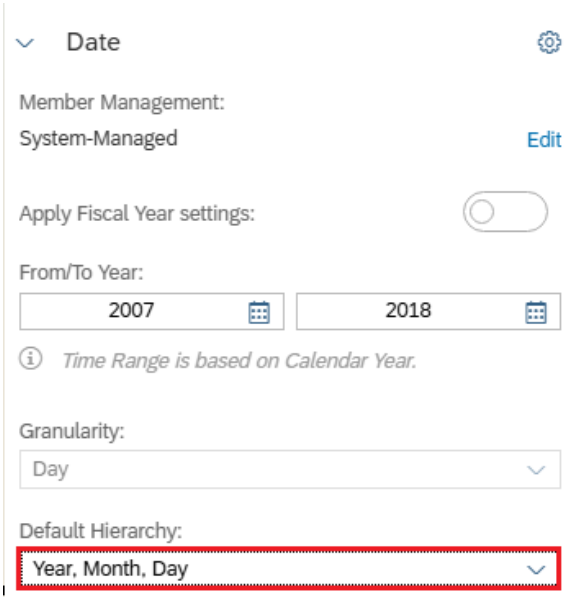
Here, under *Dimension Table*, click **Edit Dimension Table** .

Edit Dimension Table



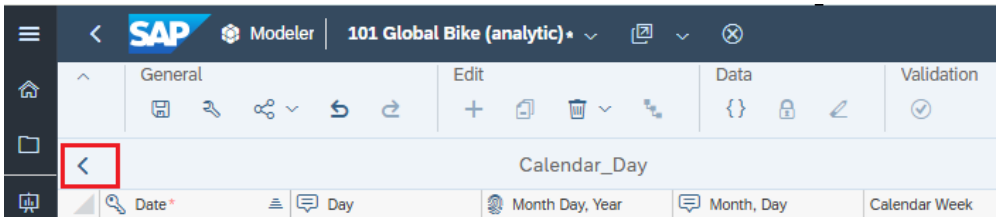
This takes you to the data table and in the right properties window for the *Dimension Table*. To obtain a granularity of days, in the *Default Hierarchy* field, choose **Year, Month, Day**.

Year, Month, Day



To return to the dimension view, choose **Back** in the header area.

Back

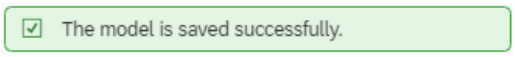


Save your model before proceeding. To do this, click **Save** **icon in the** *General* **area of** the **ribbon**.

Save

Stay in this window and continue with the next step.

The SAC system confirms the successful backup of your model with a success message.



Remain in this window for the next step.

Step 4: Change Dimension Product

Task Change the dimension *Product*. **Time** 15 min.

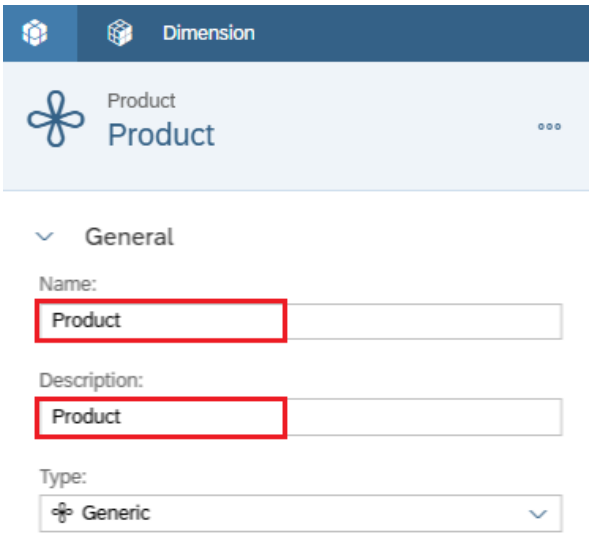
Description Change the Product dimension for your model.

Your *### Global Bike (analytic)* model should still be open. If not, restart the *Modeler* and open your model.

Initial Screen

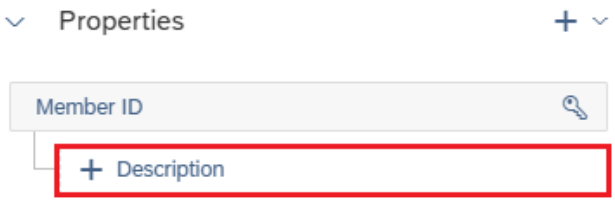
Under *Dimensions*, select the **Product_ID** dimension and change *the Name* and *Description* to **Product**.

Product_ID
Product



Under *Properties*, click **Description**. This defines one of the dimensions from your Excel file as a description field.

Description



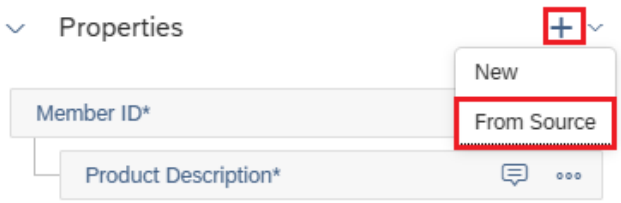
In the *Add Description Property* window, choose **Product Description** and confirm the window with **OK**.

Product Description
Ok then



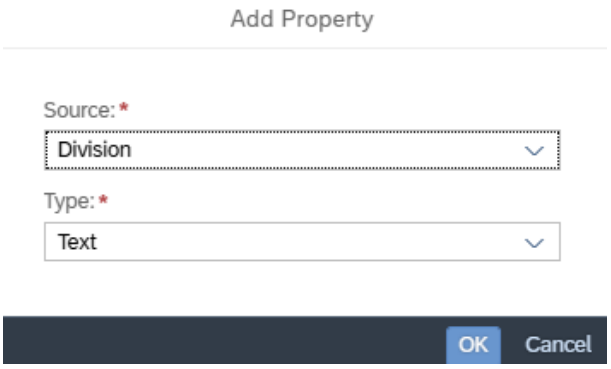
To add an attribute, click the **Add Properties** + icon in the *Properties* pane and choose **From Source**.

Add Property
From Source



In the *Add Property* window, choose **Division** in the *Source* field. Select **Text** as the *Type*. Confirm the window with **OK.**

Division
Text
Ok then



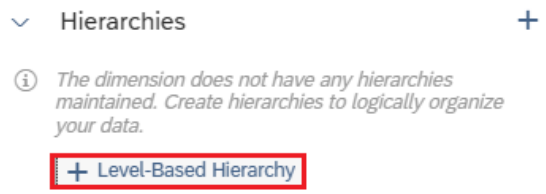
Repeat the step and define the attributes **Product Category, Color**, and **Components**. *Type* is **text** in each case. At the end, your list of *properties* should look like this.

Product Category
Color
Components
Text



Now create a hierarchy for your dimension. In the right *dimension* window, scroll to the Hierarchies section. Click **Level-Based Hierarchy**.

Level-Based Hierarchy

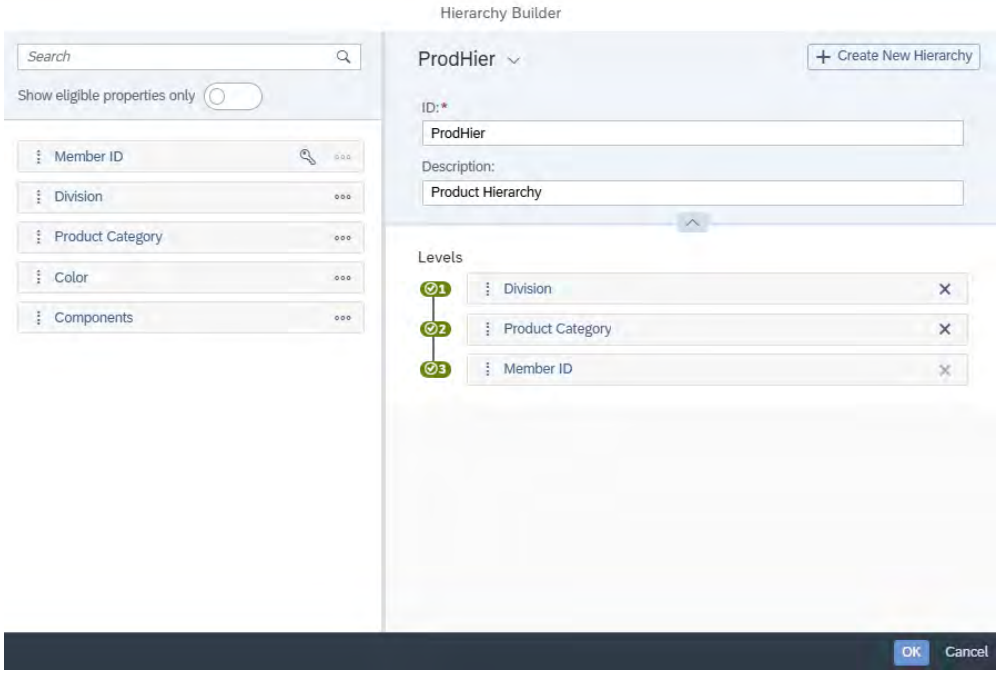


In the upper text box, enter **ProdHier** as the *ID* for your hierarchy and **Product Hierarchy** as the *description*.

ProdHier
Product Hierarchy

By dragging and dropping, choose **Product Category** and then **Division** from the list on the left and place it as follows.

Product Category
Division



Then click **OK**.

OK

Save your model before proceeding. To do this, click **Save** icon in the *General* area of the ribbon.

Save

Step 5: Change Dimension Customer

Task Change the dimension *Customer*. **Time** 15 min.

Description Change the Customer dimension for your model.

Your ### *Global Bike (analytic)* model should still be open. If not, restart the *Modeler* and open your model.

Initial Screen

Under *Dimensions*, select the **Customer_ID** dimension and change *the Name* and *Description* to **Customer**.

Customer_ID
Customer

Under *Properties*, define the **Customer_Description** field as *Description*.

Customer_Description

Create the following attributes as described in the previous step.

Source Field	Types
Country	Text
Country Text	Text
Sales Organization	Text
Region	Text
City	Text

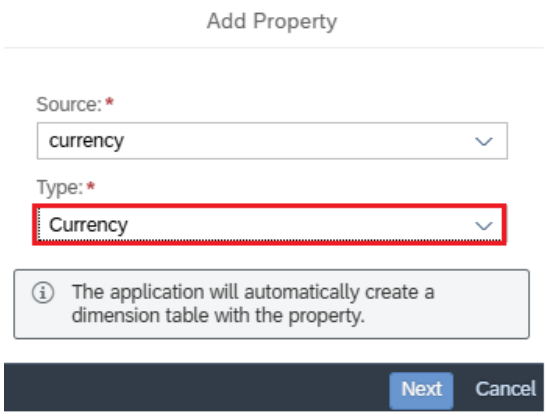
Country
Country Text
Sales Organization
Region
City

Now add the **currency** dimension as an attribute.

currency

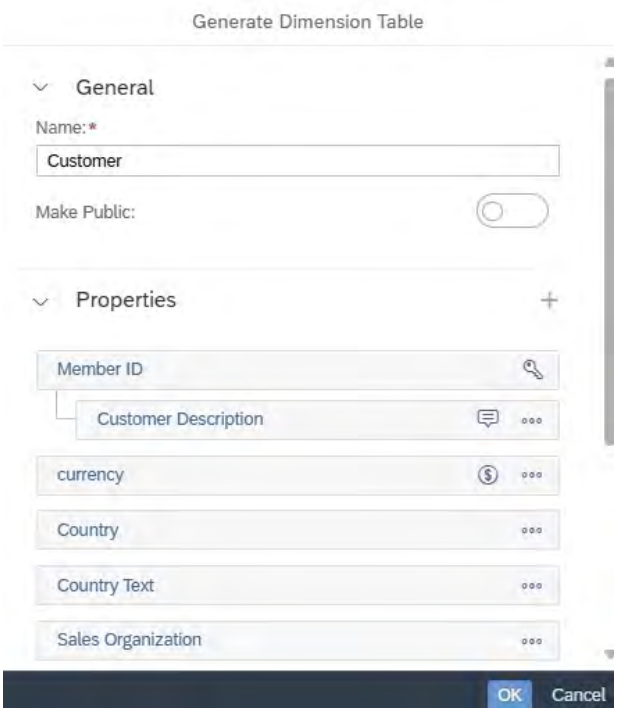
Under *Type*, select **Currency** and choose **Next**.

Currency
Next



Confirm the *Generate Dimension Table* window with **OK**.

Ok then



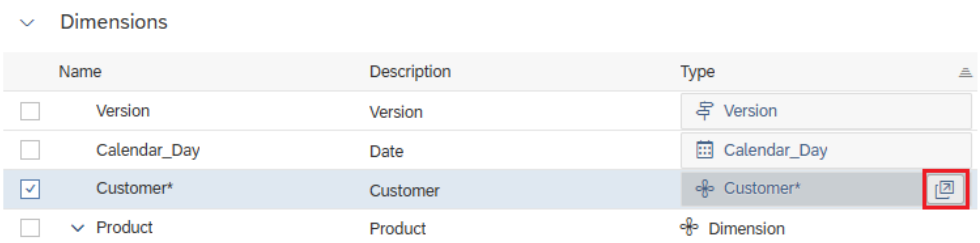
Now your list of *properties* should look like this.



Since *currency* is created as a system property, each value MUST have a currency. The system automatically creates the proficiency #. A currency must also be assigned to this.

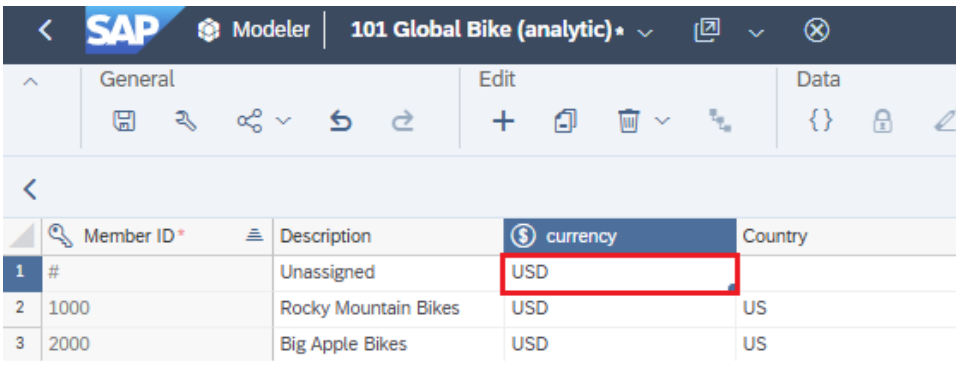
In the list of dimensions, go to the *Customer type* and choose **Edit Dimension Table** .

Edit Dimension Table



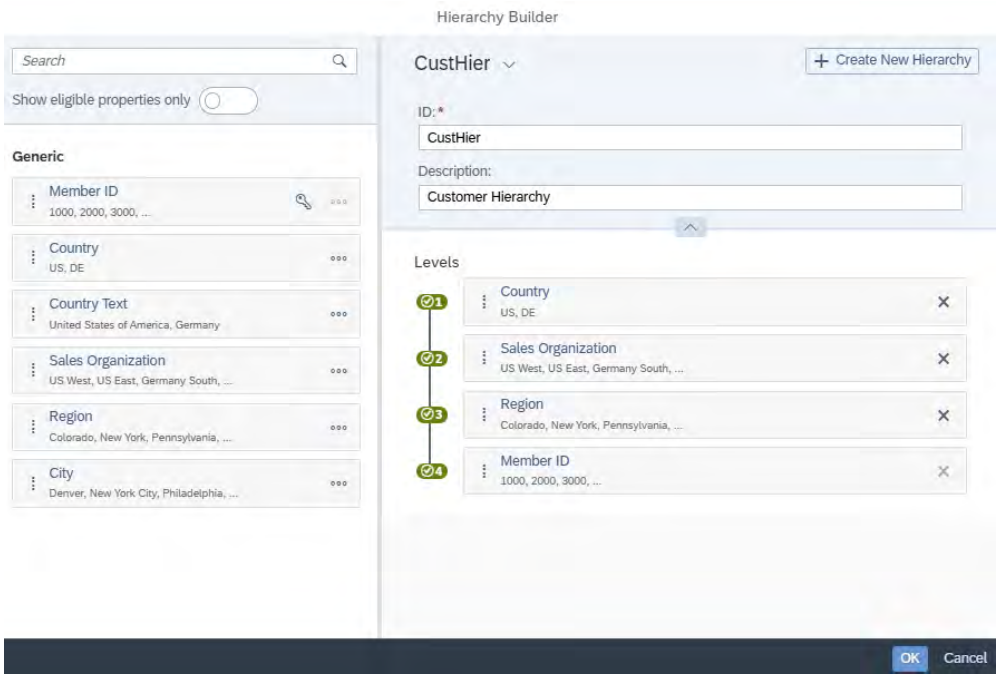
For entry #, enter **USD** under *currency*.

USD



Now create the following hierarchy for your Customer dimension. In the right *dimension* window, scroll to the Hierarchies section. Click Create **Hierarchy** and choose **Level-Based Hierarchy**.

Create Hierarchy
Level-Based Hierarchy



Then click OK

Ok then

To return to the dimension view, choose **Back** in the header area.

Back

Save your model before proceeding. To do this, click **Save** icon in the *General* area of the ribbon.

Save

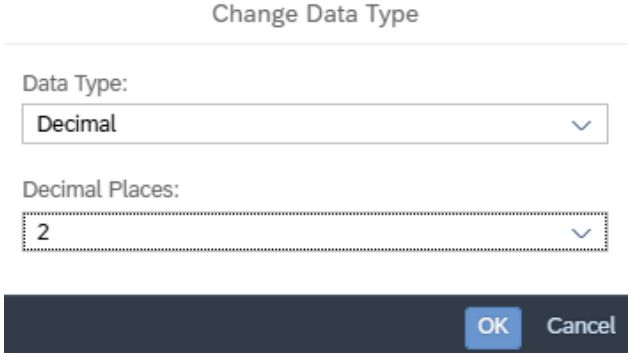
Question Does a city hierarchy level make sense in the Customer dimension?

☐

Step 6: Change key figures

Task Change the settings of the key figures. **Time** 15 min.

Description Change the six key figures and create a calculated key figure.

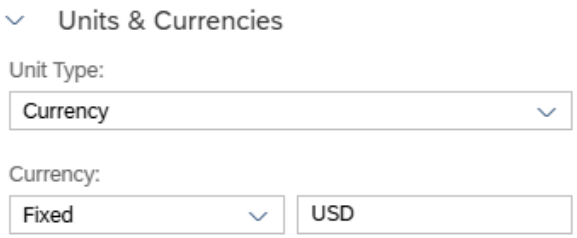
Your ### *Global Bike (analytic)* model should still be open. If not, restart the *Modeler* and open your model.	Initial Screen
Under *Measures*, select the measure Revenue_GC by clicking it.	Revenue_GC
Note Models in financial and business reporting can have different measures that are reported in both global currency (GC) and local currency (LC). Revenue GC (Global Currency) and Revenue LC (Local Currency) are terms used in financial reporting and in the international business environment. They refer to the display of sales in different currencies. **Revenue GC (Global Currency):** Refers to the recording and reporting of sales in a single, global currency, often referred to as consolidation currency. This currency is used to standardize the financial performance of a multinational company so that financial information is comparable across different countries and regions. A strong and stable currency such as the US dollar or the euro is often used for this purpose. The conversion is done using exchange rates to transfer local sales to this global currency. **Revenue LC (Local Currency):** This key figure shows the revenue as recorded in the respective national currency of the location where the company operates. Sales figures in local currency provide an accurate view of financial performance in a specific market or country without the influence of exchange rate fluctuations. By using Revenue GC and Revenue LC, companies can both analyze local market performance and get a consolidated view of the entire organization, which is essential for strategic decisions and reporting.	
In the *Data Type* area, select *Decimal (7)* by clicking on it. In the *Change Data Type* window, choose 2 under *Decimal Places* and confirm the window with **OK.** 	2 OK

In the *Aggregation* section, in the *Aggregation Type* field, choose **Sum**. Leave all other fields in this area empty.

Sum

In the *Units & Currencies* area, choose **Currency** in the *Unit Type* field. You want to use US dollars as the fixed currency for this key figure. Under *Currency*, choose **Fixed** and check that **USD** is entered as the currency.

Currency
Fixed
USD



Under *Measures*, select the measure **Revenue_LC** by clicking it.

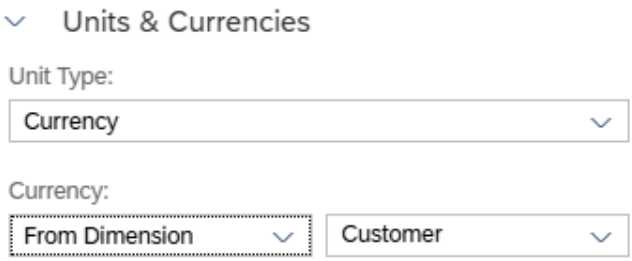
Revenue_LC

For the *data type*, change the *Decimal Places* to 2 and set **sum** for the *aggregation type*.

Decimal 2
SUM

In the *Units & Currencies* area, choose **Currency** in the *Unit Type* field. For this key figure, the respective currency from the *Customer* dimension should be used instead of the US dollar as the fixed currency. To do this, choose *Currency* **From Dimension** in the field and check that the dimension **Customer** has been entered.

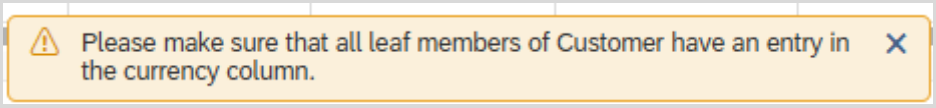
Currency
From Dimension
Customer



Adjust the following key figures in this way.

Name	Data Type	Decimal Places	Aggregation Type	Unit Type	Unit/ Currency
Discount_GC	Decimal	2	Sum	Currency	Fixed USD
Discount_LC	Decimal	2	Sum	Currency	From Dimension *Customer*
Costs_GC	Decimal	2	Sum	Currency	Fixed USD

Note You have certainly noticed the following system hint whenever you select a currency unit with reference to a dimension.



A currency unit can only be used in the key figure if a currency unit is maintained for each *customer* (that is, each entry in the customer dimension).

Save

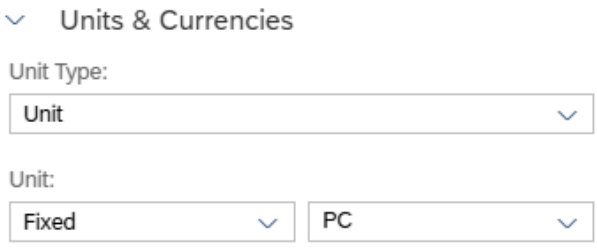
To save your previous entries, click **Save** icon in the *General* area of the ribbon.

Sales_Quantity
Integer

Up to now, you have changed five currency key figures. Now adjust the key figure **Sales_Quantity**. Make sure that **integer** is set as *the data type*.

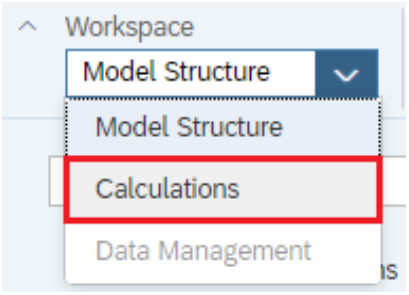
Sum
Unit
PC

In the *Aggregations* area, choose Aggregation *Type* **Sum**. In the *Units & Currencies* area, select **Unit** as *the Unit Type*. In the new field, enter **PC** next to *Fixed*.



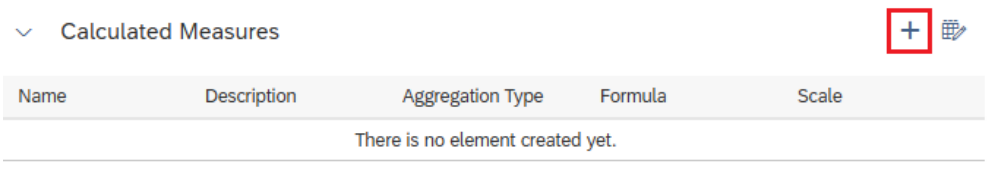
Calculations

Finally, you create a calculated key figure. To do so, switch to **Calculations** in the upper left of the *Workspace* area.



Add Calculated Measure

In the *Calculated Measures* section, click **Add Calculated Measure** + .



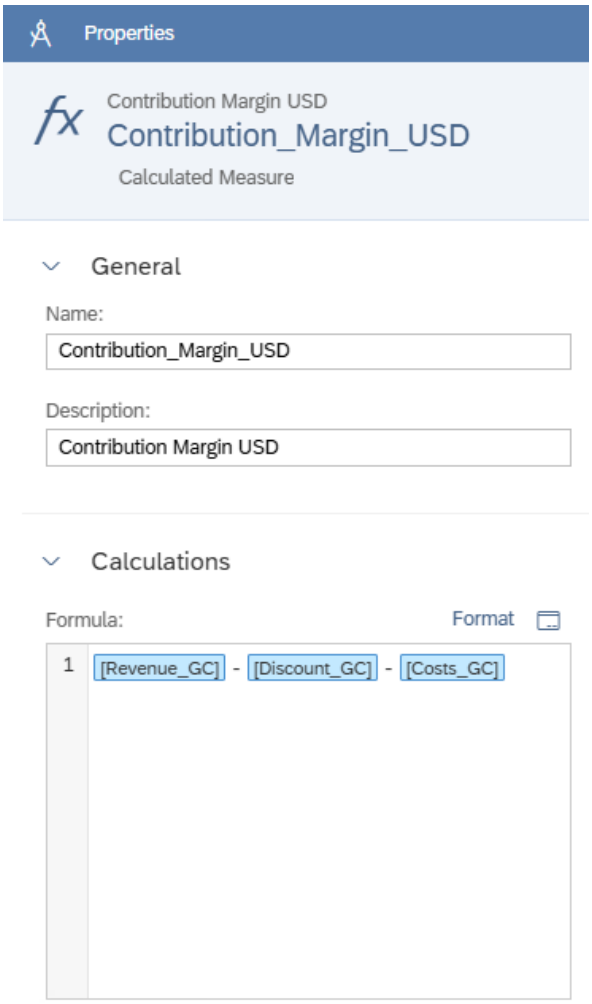
ContributionMarginUSD
Contribution Margin USD

In the *properties* window on the right, enter **Contribution_Margin_USD** as *the name* and **CONTRIBUTION MARGIN USD** as the *description*.

Currency
USD

In the *Units & Currencies* area, in the *Unit Type* field, select **Currency** and verify that **USD** is filled in as the currency.

Now turn to the formula window in the middle. Enter the following formula. Each time you start typing, the system suggests functions (in capital letters) and key figures (in square brackets). In your case, select the right metrics with a simple click.



After you have checked the formula again, save your model by clicking **Save** icon in the *General* area of the ribbon.

Save

Step 7: Load Transaction Data 2019

Task Additionally load the transaction data for the year 2019 into your model. **Time** 15 min.

Description Load more sales data into your analytic model annually.

Your ### *Global Bike (analytic)* model should still be open. If not, restart the *Modeler* and open your model.

Initial Screen

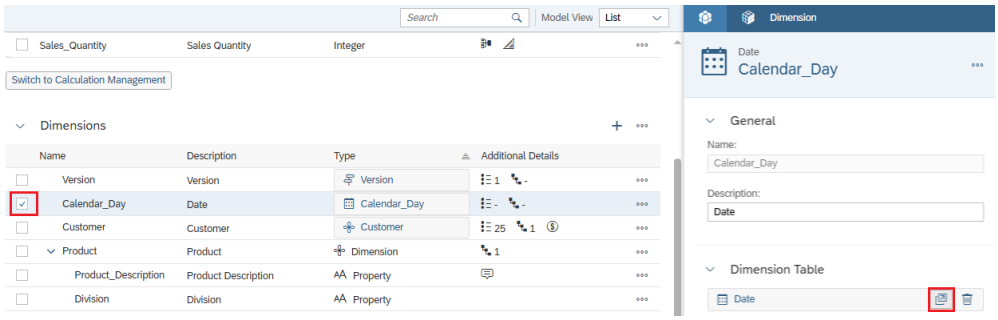
In this step, load past sales figures into the fact table of your model.

Before you can extend the data model with the 2019 sales figures, you need to extend the time dimension by the year 2019.

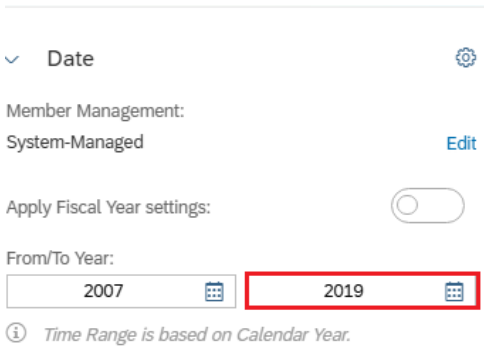
Note The time dimension can also be extended in advance for the coming years.

To do so, choose Calendar_Day under Dimensions in the Model Structure pane and navigate to **Edit Dimension Table**.

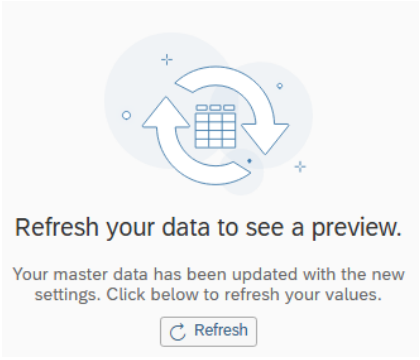
Calendar_Day
Edit Dimension Table



Under *Date*, extend the time dimension to include the year 2019.



You then see the following message:



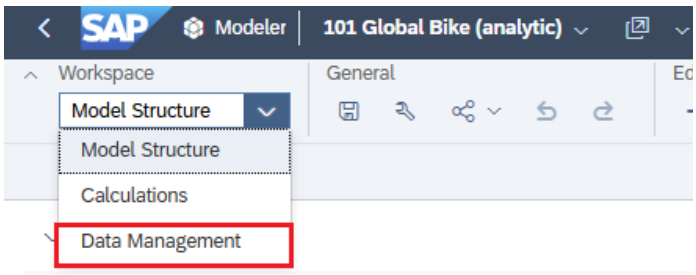
Click **Refresh** to see the previous data again.

Refresh

The transaction data for 2019 must now be loaded.

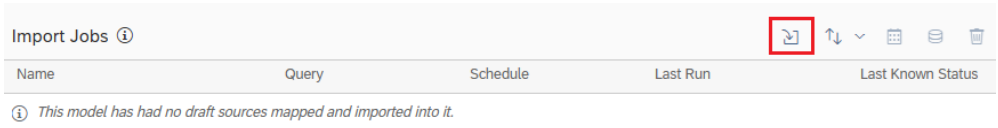
To do this, switch to Data Management in the *Workplace* area on the upper left**.**

Data Management



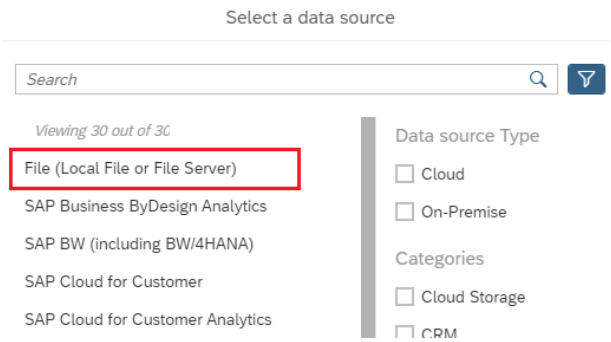
For *import jobs*, click **Import Data** and choose.

Import Data



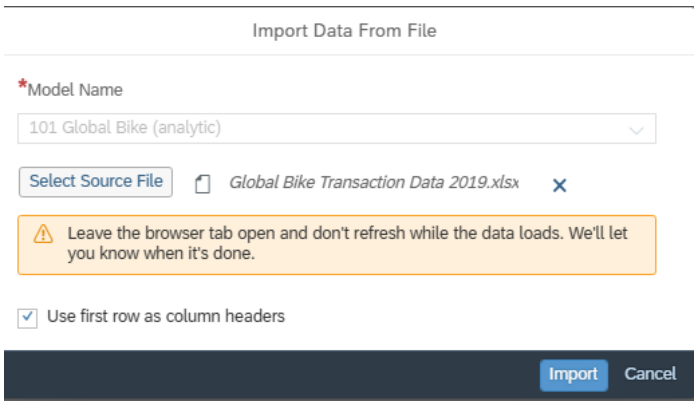
Choose **File** (Local File or File Server).

File



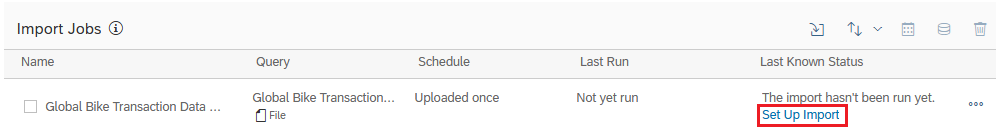
In the window that opens, click *Select Source File.* Search for the Excel file *Global Bike Transaction Data 2019.xlsx* and choose Import.

Import



After the file is uploaded, choose **Set Up Import**.

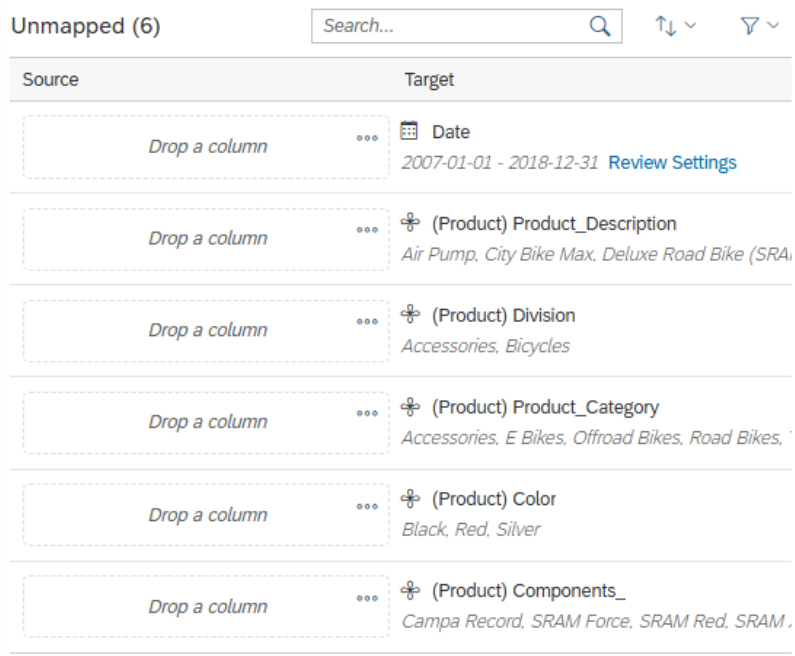
Set Up Import



In the following view, choose Next at the bottom right.

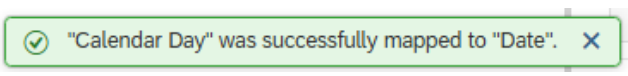
Next

On the following screen, on the right, you can see that 9 facts could already be *mapped* automatically. In the middle of the window, you can see that 6 facts were not found automatically (*unmapped*).



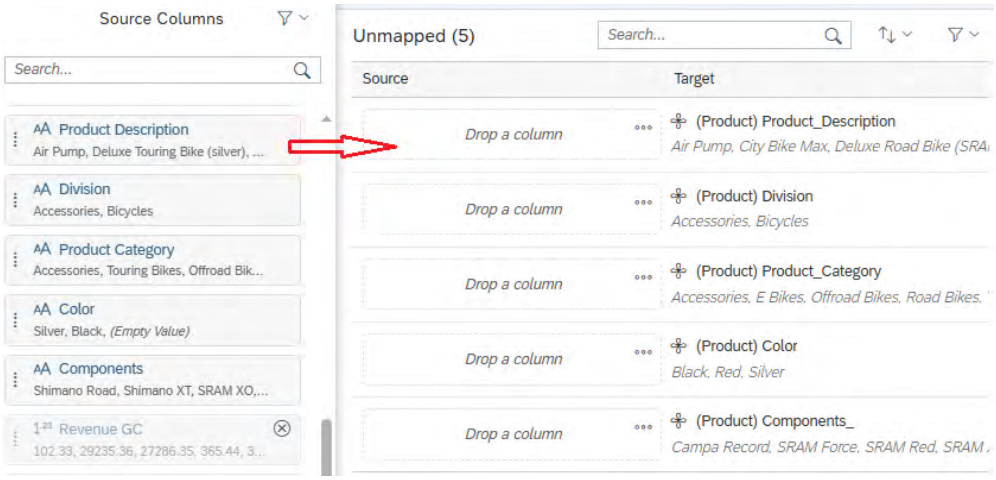
On the left-hand side, in the *Source Columns*, find **Calendar Day**. Use drag and drop to drag it to the *Drag a column* area for the entry *Date*.

Calendar Day

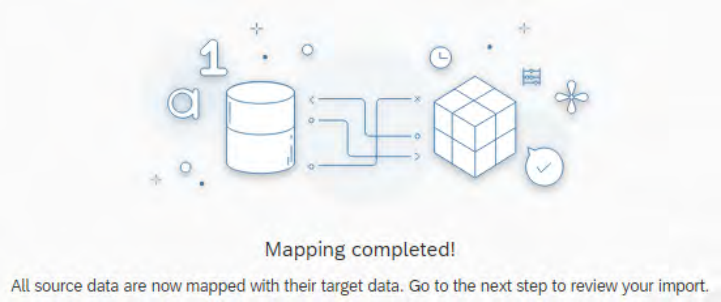


In the same way, assign **Product Description, Division, Product Category, Color, and Components** to the Target **Unmapped (5)**.

Product Description, Division, Product Category, Color, and Components

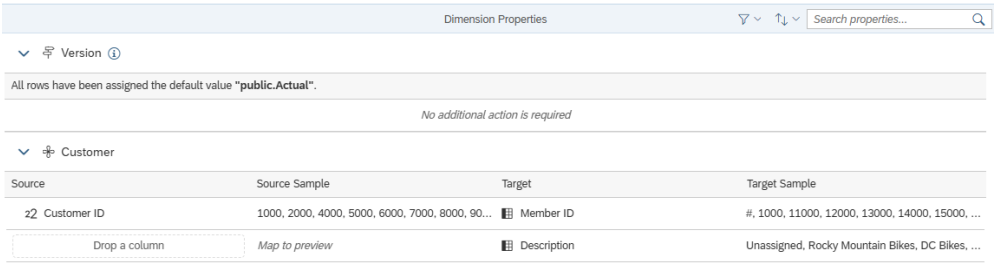


You should now see the following message in the middle of the screen.



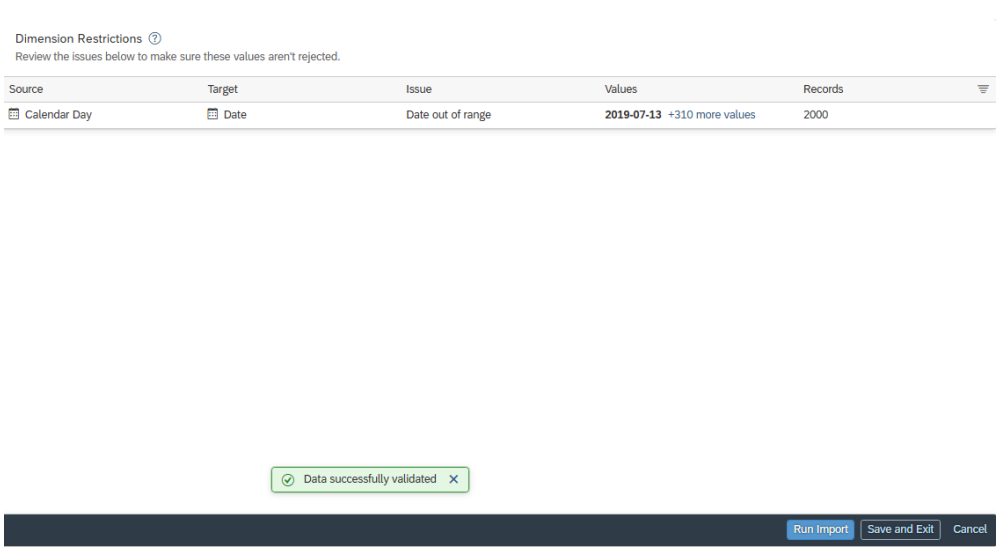
Click Next in the bottom-right corner. You will receive this notification.

Next

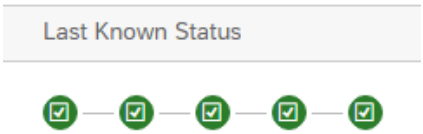


Click on Next again and you will receive the following success message.

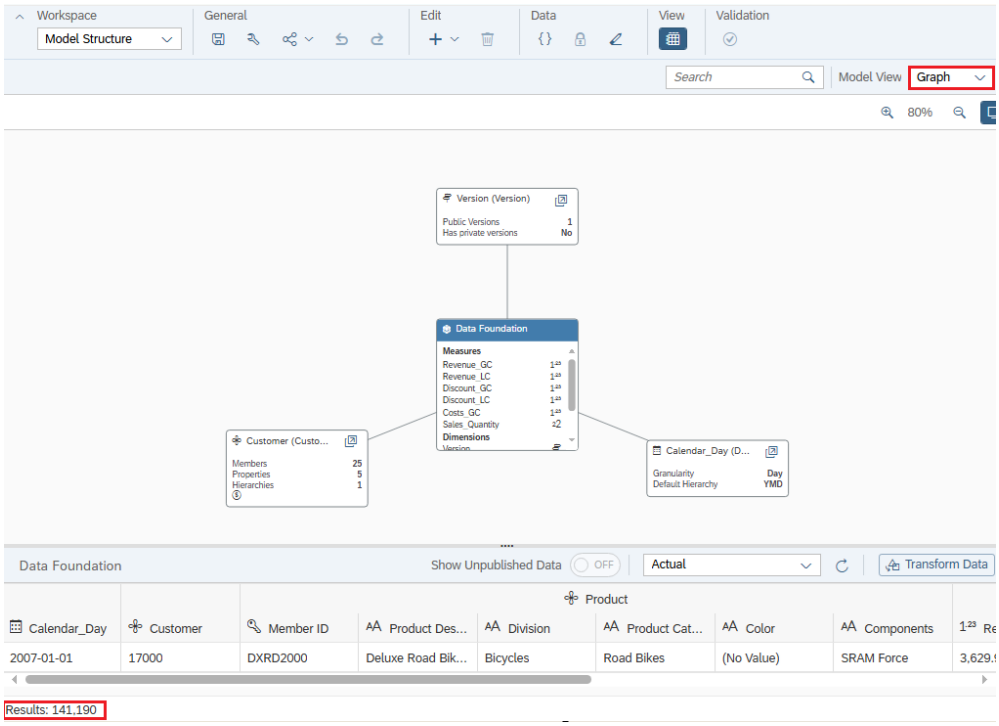
Next



Now click on Run Import on the bottom right and then on Finish. As a result, you see the following status display.



To return to your model, go back to *Model Structure* in the *Workplace* area. Your final model should look like this.



Under your model, you can see in the preview window that a total of 141,190 rows have been loaded.

Save your model by clicking **Save**  in the top menu bar. It is now ready for use.

Save

Step 8: Create story

Task Create a simple story. **Time** 30 min.

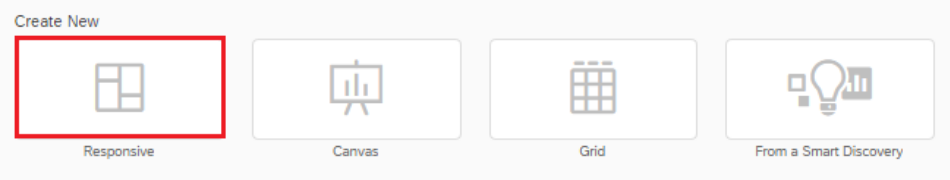
Description Create a story based on your new model to test it.

Initial Screen

To create a new story, open the *Story Builder* in the left menu bar by clicking  (*Stories*).

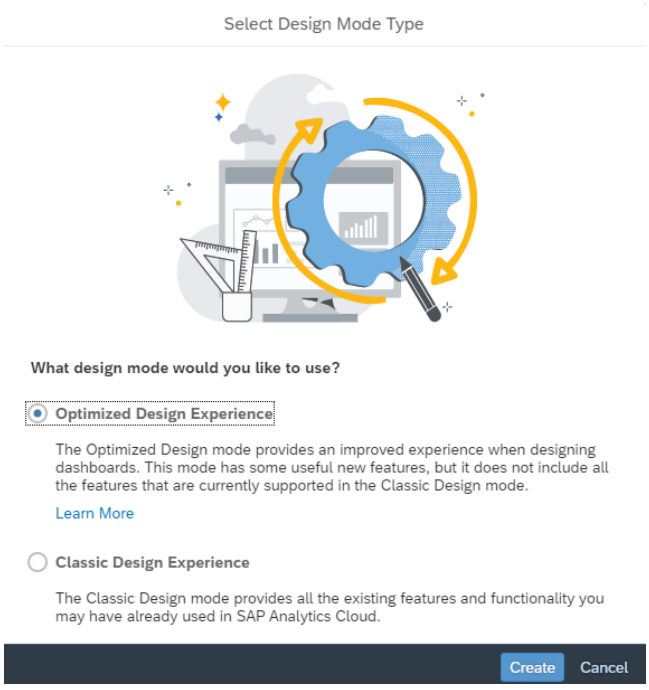
Responsive

In the *Create New* area, choose *Responsive*.



Optimized Design Experience

In the window that opens, *choose Select Design Mode Type* **Optimized Design Experience** and **Create**.



Under View, click left side panel and close it and open the right side panel**.**



Chart

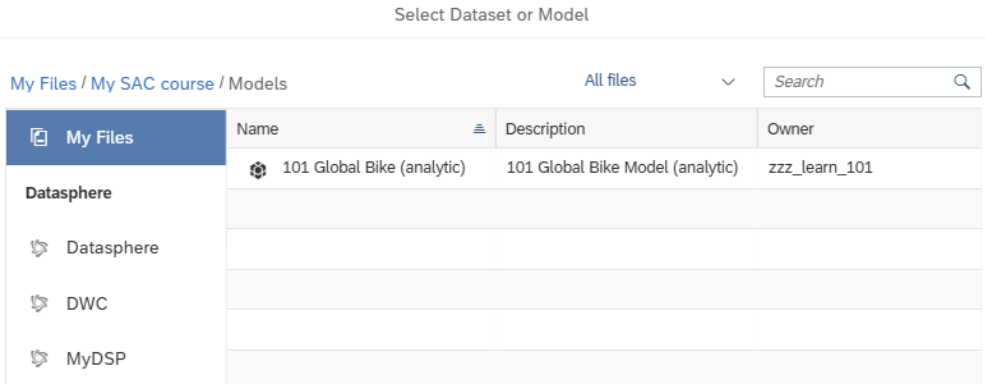
To embed a graphic as an object type in your story first, choose the **Chart** icon in the *Insert* area in the upper menu bar.



You now need to define which data source is to be used. Navigate to your *My SAC course* folder and then to the *Models* subfolder. Choose your Model ### **Global Bike (analytic)**.

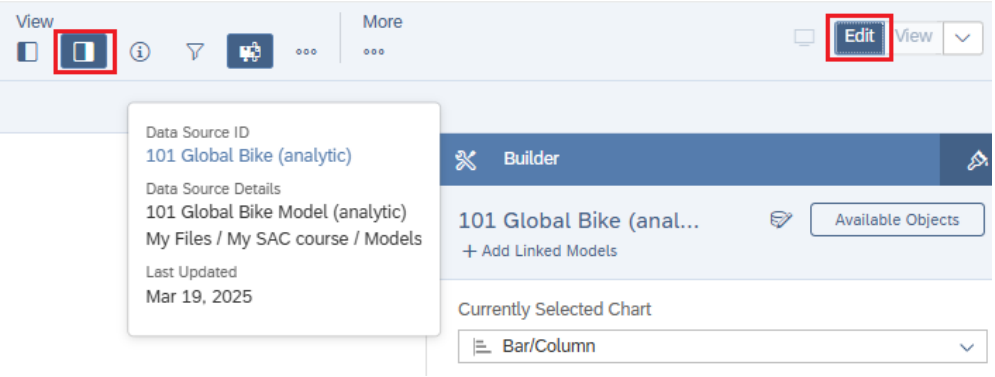
Global Bike (analytic)

Existing Model



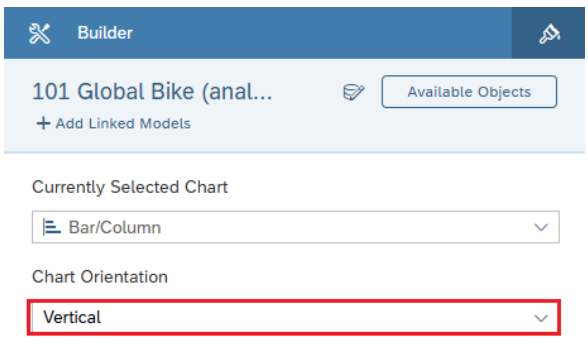
Make sure you are in edit mode and have the Right Side Panel open.

Edit Mode
Right Side Panel



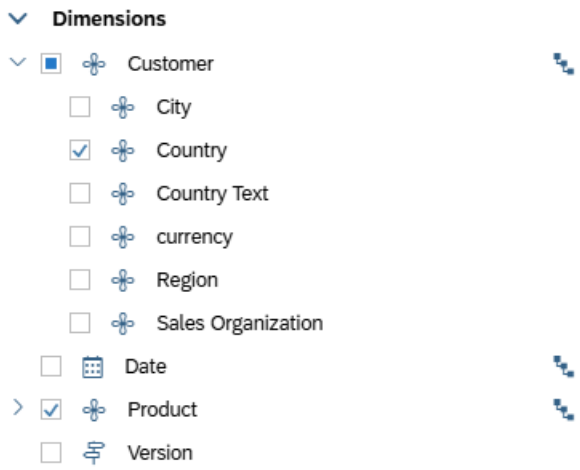
Change the orientation of the chart to *Vertical*.

Vertical



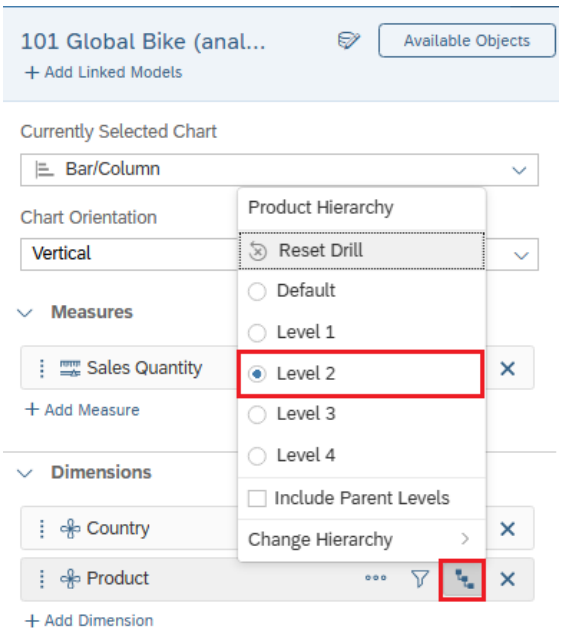
In the *Measures* section, add the measure **Sales Quantity**, and in the *Dimensions* section, the dimensions **Country** (under *Customer*) and **Product**.

Sales Quantity
Country
Product



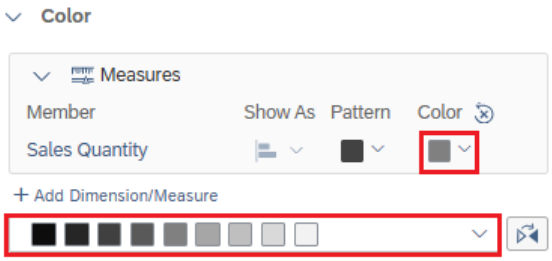
For the dimension *Product*, set the hierarchy level to *level 2* by hovering over the dimension *Product* and clicking the icon that appears. Select **Level 2** here.

Level 2



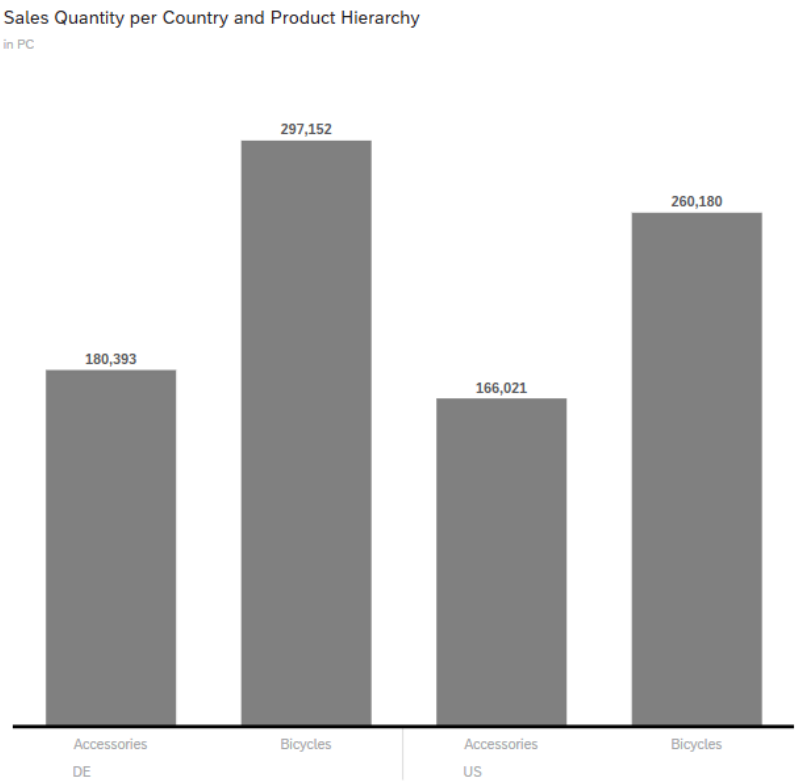
Change the gradient to **grayscale** and choose a **gray tone** as *the color*.

Grayscale, grey tone



Increase the size of the chart approximately to twice the size. Change the title to **Sales Quantity per Country and Product Hierarchy** by double-clicking the chart header.

Sales Quantity per Country and Product Hierarchy



Insert a new area next to the graphic. To do this, click the area next to the graphic and choose **More Actions** ► **Add Lane** ► **Add Lane to Right.**

More Actions → Add Lane → Add Lane to Right

In the new area, insert a table using the **Table** icon.

Table

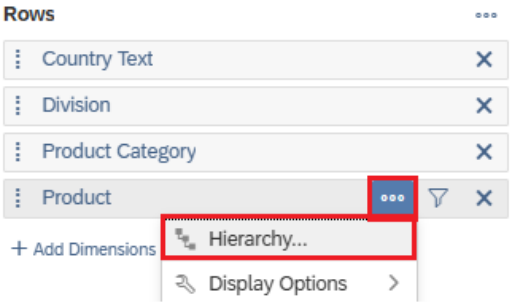
Under *Rows*, add the dimensions **Country Text**, **Division**, **Product Category**, and **Product**, and under *Columns*, add the key figures **Sales Quantity**, **Revenue LC**, and **Discount LC**.

Country Text, Division, Product Category, Product
Sales quantity
Revenue LC, Discount LC

Note The order of the checkbox of the boxes determines the order of the key figures in the table.

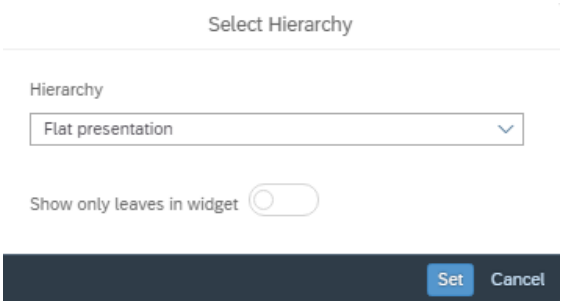
For *Product*, switch from the hierarchical display to the flat display. To do so, in the *Builder* panel on the *right side* panel, under *Measures*, click **More** and choose **Hierarchy….**

More
Hierarchy…



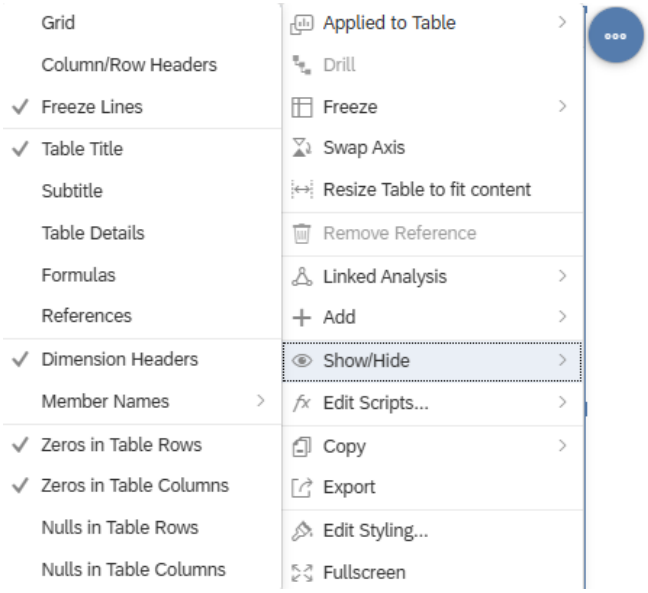
In the window that opens, under *Hierarchy*, select the **Flat Presentation** option and close the window by choosing **Set**.

Flat Presentation
Set



Next, remove the table title, subtitle, and table details by clicking your table and choosing **More Actions** in the upper right corner. In the menu, choose **Show/Hide**. In the following submenu, deselect **Subtitle** and **Table Details**.

Show/Hide
~~Subtitle~~
~~Table Details~~



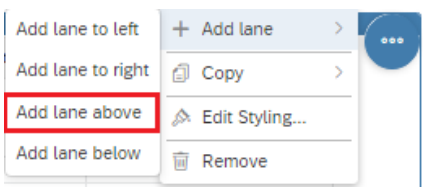
Rename your table to **Detail Product Analysis**.

Detailed Product Analysis

Now enlarge the area of the *table* using the left side. The table should be visible without horizontal scrollbars.

For filters, insert a new area above the graphic and the table. To do this, click on the area of the graphic and choose **More Actions** ▶ **Add Lane** ▶ **Add Lane Above.**

More Actions → Add Lane → Add Lane Above



Define a *Page Filter* on *Date* that allows end users to select a maximum of the last 5 years.

To do this, click in the new area and insert a filter using the *Insert* ribbon and click on the icon **input control** .

Input Control



Dimensions →
Date →
Filter by Member…

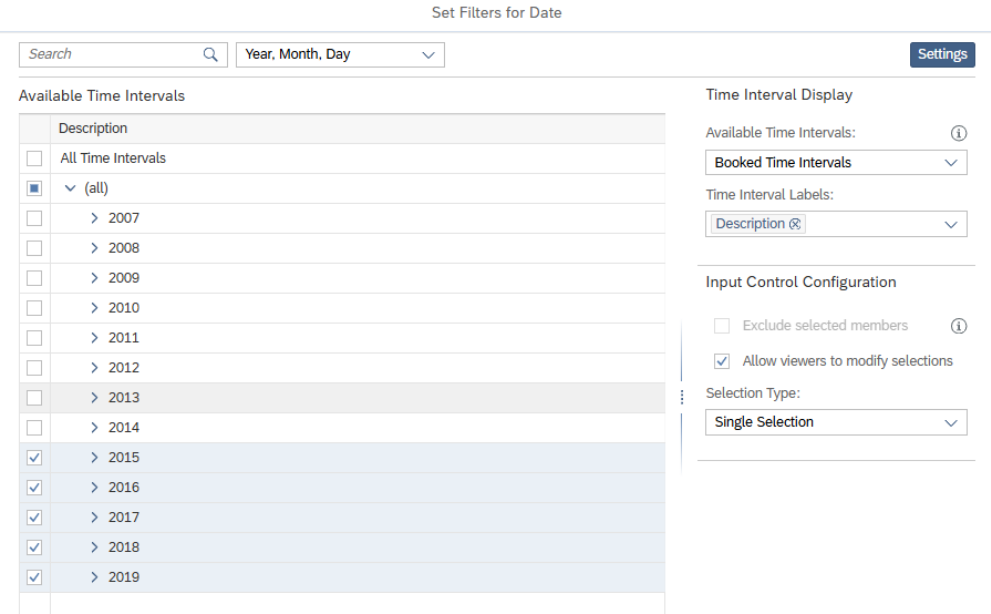
The system then inserts a *Page Filter* element. You must now define this. To do this, select the entry **Dimensions ► Date ► Filter by Member…** in the menu that opens.

In the following dialog, select the values for **2015 - 2019** individually.

2015 - 2019

Since only a single selection of the year is to be possible in this filter, choose the **Settings** button and define the *selection type* as **Single Selection**. Confirm the window with **OK.**

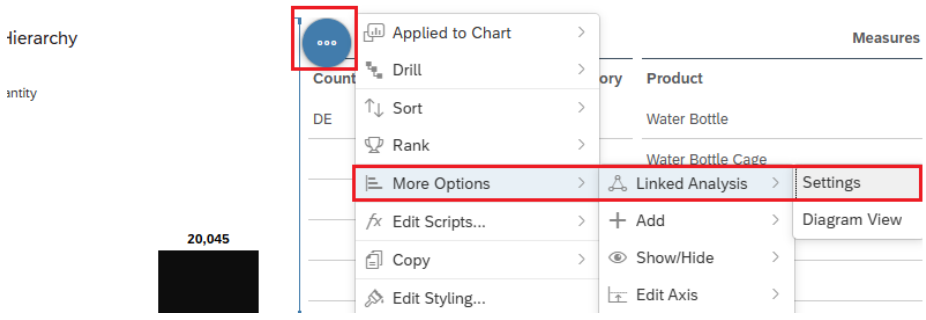
Settings
Single Selection
Ok then



Define a linked analysis in the form that a selection of a graphical element filters the table.

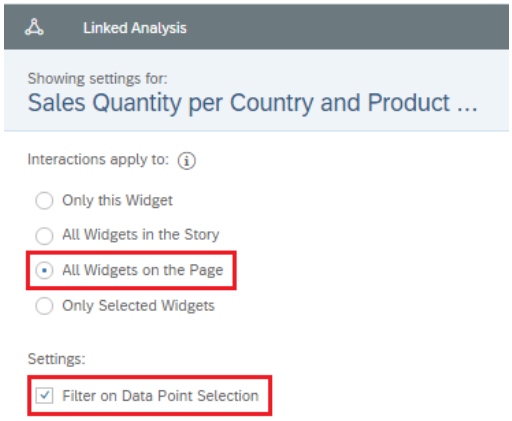
When you click on the graphic, the option **More Actions** appears in the upper right corner. Click the button. In the context menu, choose **More Options ► Linked Analysis ► Settings**.

More Actions
More Options →
Linked Analysis →
Settings



The definition view now opens in the right-hand pane. First, in the *Interactions apply to* area, select **All Widgets on the Page**. In the *Settings* area that then appears, choose the **Filter on Data Point Selection** option and close the window by **choosing Apply**.

All Widgets on the Page
Filter on Data Point Selection
Apply

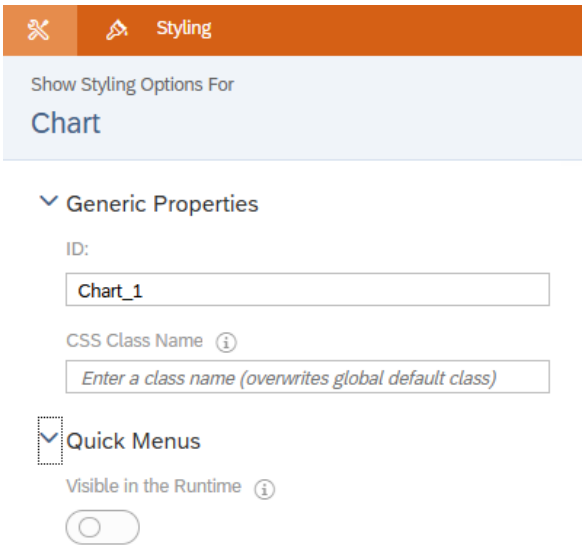


In the *right-hand side panel*, switch from the *Builder* panel to the **Styling** panel.

Styling

In the *Quick Menus* area of the *right-hand side panel*, deselect it for the *graphic*, *table*, and *page filter*.

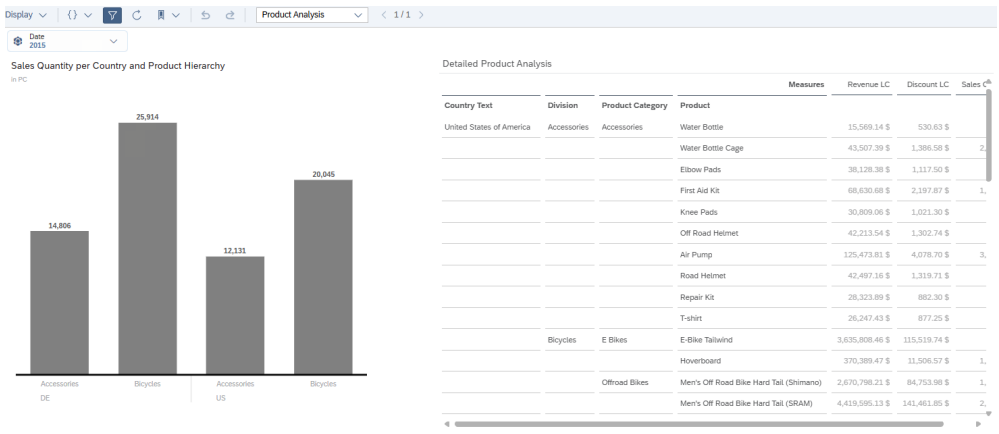
~~Quick Menu~~



Rename your page from Page_1 to **Product Analysis**.

Product Analysis

Your story should now look like this.



Finally, save your story by clicking **Save** in the upper left corner. In the window that opens, choose your personal folder *My SAC course* as the location, go to the *stories* subfolder, and name your story ### **Story 3**.

Save
Story 3

Challenge 1: Calculated Measures in Model or Story

Challenge Create a new calculated measure *Discount %*. **Time** 15 min.

Description Create the measure twice, in the model and in the story, and examine the differences between the two.

Initial Screen

A calculated measure can be created either directly in the model or in a story built on top of it.

Model

In the model: In this case study, you have already created the calculated measure *Contribution Margin* in your *### Global Bike (analytic)* model.

Story

In the story: in the SAC Story Builder exercise, you have already used the calculated measure *Discount %*. In the Story Builder case study, you have created a new calculated measure *contribution margin %*.

Discount %

In your *### Global Bike (analytic)* model, create a new calculated key figure *Discount %* with the correct formula.

You then create the same calculated measure in your story **### Story 3**.

Question When does it make more sense to create the calculated measure in the model and when in the story?

Note that, depending on the use case, one or more stories can be based on the same model.